Determination of total aluminum, chromium, copper, iron, manganese, and nickel and their fractions leached to the infusions of black tea, green tea...

Total aluminum, chromium, copper, iron, manganese, and nickel were determined in black tea, green tea, Hibiscus sabdariffa, and Ilex paraguariensis (mate) by electrothermal atomic absorption spectrometry after nitric/perchloric acid digestion. In each case, one ground sample of commercially available leafy material was prepared and three 0.5-g subsamples were run in parallel. The infusions were also analyzed and the percentage of each element leached into the liquor was evaluated. The obtained results indicated that hibiscus and mate contained lower levels of aluminum (272+/-19 microg/g and 369+/-22 microg/g, respectively) as referred to black tea (759+/-31 microg/g) or green tea (919micro29 microg/g) and suggested that mate drinking could be a good dietary source of essential micronutrient manganese (total content 2223+/-110 microg/g, 48.1% leached to the infusion). It was also found that the infusion of hibiscus could supply greater amounts of iron (111+/-5 microg/g total, 40.5% leached) and copper (5.9+/-0.3 microg/g total, 93.4% leached) as compared to other infusions. Moreover, it was found that the percentage of element leached to the infusion was strongly related to the tannins content in the beverage (correlation coefficients > 0.82 with the exception for nickel); for lower tannins level, better leaching was observed.